1. A subset of size four (4) is to be chosen from

$$\{B, D, A, E, C, F, G, H\},$$

with all subsets of size 4 equally likely. Use the sequential random-subset procedure discussed in class.

(a) (2 marks) Suppose the random numbers generated from Unif$(0, 1)$, in order, are

$$0.52, \quad 0.57, \quad 0.66, \quad 0.30, \quad 0.80, \quad 0.90, \quad 0.25, \quad 0.70.$$

Determine the subset selected. (No partial credit).

Solution: The first six decisions are

Item	Random number	Selection probability	Decision
B	0.52	$\frac{4}{8}$	Reject
D	0.57	$\frac{4}{7}$	Select
A	0.66	$\frac{3}{6}$	Reject
E	0.30	$\frac{3}{5}$	Select
C	0.80	$\frac{2}{4}$	Reject
F	0.90	$\frac{2}{3}$	Reject

At this point, two more elements are required and only G and H remain. Both must be selected, and the last two random numbers are not used.

Hence, the selected subset is

$$\boxed{\{D, E, G, H\}}.$$

(b) (4 marks) Construct the decision path that results in the subset

$$\{D, E, F, H\}.$$

At each step, show the boundary value, the open, closed, or semi-open interval of random numbers that provides the required decision, and the decision, in a table of the form

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Item | Boundary Value | Required Interval  | Decision
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Solution: The required decision path is

Item	Extremal boundary	Required range	Decision
B	$\frac{1}{2}$	$U_1 \geq \frac{1}{2}$	Reject
D	$\frac{4}{7}$	$U_2 < \frac{4}{7}$	Select
A	$\frac{1}{2}$	$U_3 \geq \frac{1}{2}$	Reject
E	$\frac{3}{5}$	$U_4 < \frac{3}{5}$	Select
C	$\frac{1}{2}$	$U_5 \geq \frac{1}{2}$	Reject
F	$\frac{2}{3}$	$U_6 < \frac{2}{3}$	Select
G	$\frac{1}{2}$	$U_7 \geq \frac{1}{2}$	Reject
H	1	forced	Select

Hence the selected subset is

$$\boxed{\{D, E, F, H\}}.$$

2. CS 215 has 44 scheduled classes this semester, excluding optional help sessions. Per IITB rules, 100% attendance is expected, but a student who attends at least 80% of the classes cannot receive a DX grade for attendance (although they can fail the course for other valid reasons). After 21 classes, the attendance record is

Student	Classes attended
A	16
B	18
C	20

Due to other distractions, students A, B, and C consider probabilistic attendance, i.e., gambling on the attendance requirement, for the rest of the semester. Of the 23 remaining classes, each plans to set aside 2 classes as a buffer for unexpected events, and will attend these 2 for certain. For each of the other classes, the student independently attends with probability 0.75.

Each student is willing to take this gamble only if the probability of falling below the 80% attendance threshold is at most 4%.

(a) (1 mark) Identify the Binomial parameters and complete this statement
$X \sim \text{Binomial}(_____, _____)$.

Solution: The correct binomial model is $X \sim \text{Binomial}(21, 0.75)$.

(b) (6 marks) The following probabilities are provided.

x	13	14	15	16	17	18	19	20	21
$P(X \leq x)$	0.130	0.256	0.433	0.633	0.808	0.925	0.981	0.998	1.000

For each student, show the relevant probability calculation and state whether the student takes the gamble.

A. B. C.

Solution: A student must attend at least 80% of the 44 classes. Since

$$0.8(44) = 35.2,$$

at least 36 classes must be attended. Each student will attend 2 of the remaining classes for certain. Let X denote the number attended among the other 21 classes. Then

$$X \sim \text{Binomial}(21, 0.75).$$

since attending less than 21 classes is a "gamble".

1. Student A has already attended 16 classes and will attend another 2 for certain. Thus, at least
$$36 - 16 - 2 = 18$$
of the other 21 classes must be attended. Hence,
$$\begin{aligned} P(A \text{ falls below the threshold}) &= P(X \leq 17) \\ &= \boxed{0.808}. \end{aligned}$$

2. Student B needs to attend at least
$$36 - 18 - 2 = 16$$
of the other 21 classes. Therefore,
$$\begin{aligned} P(B \text{ falls below the threshold}) &= P(X \leq 15) \\ &= \boxed{0.433}. \end{aligned}$$

3. Student C needs to attend at least

$$36 - 20 - 2 = 14$$

of the other 21 classes. Therefore,

$$\begin{aligned} P(C \text{ falls below the threshold}) &= P(X \leq 13) \\ &= \boxed{0.130}. \end{aligned}$$

Since

$$0.808 > 0.04, \qquad 0.433 > 0.04, \qquad 0.130 > 0.04,$$

all three students have a probability of falling below the attendance threshold that exceeds their acceptable limit.

Hence,

$$\boxed{\text{All three students decide against the gamble.}}$$

3. One of two urns, A or B, is chosen with equal probability. Urn A contains 3 red balls and 1 blue ball, while urn B contains 1 red ball and 3 blue balls. Balls are drawn independently with replacement from the chosen urn.

(a) (3 marks) If the first two balls drawn are red, find the probability that urn A was chosen.

Solution: Let E be the event that the first two draws are red.

Then

$$P(E \mid A) = \left(\frac{3}{4}\right)^2 = \frac{9}{16},$$

while

$$P(E \mid B) = \left(\frac{1}{4}\right)^2 = \frac{1}{16}.$$

Using Bayes' theorem, and noting the prior probability of 0.5 (this is important)

$$\begin{aligned} P(A \mid E) &= \frac{P(E \mid A)P(A)}{P(E \mid A)P(A) + P(E \mid B)P(B)} \\ &= \frac{\frac{9}{16} \cdot \frac{1}{2}}{\frac{9}{16} \cdot \frac{1}{2} + \frac{1}{16} \cdot \frac{1}{2}} \\ &= \boxed{\frac{9}{10}}. \end{aligned}$$

(b) (4 marks) Suppose that among the first n draws, exactly r are red. Find the condition on r and n under which urn A is more likely than urn B.

Solution: Let E_r be the event that exactly r of the first n draws are red.

If urn A was chosen,

$$P(E_r \mid A) = \binom{n}{r} \left(\frac{3}{4}\right)^r \left(\frac{1}{4}\right)^{n-r}.$$

Similarly,

$$P(E_r \mid B) = \binom{n}{r} \left(\frac{1}{4}\right)^r \left(\frac{3}{4}\right)^{n-r}.$$

Since the two urns have equal prior probabilities,

$$P(A \mid E_r) = \frac{P(E_r \mid A)}{P(E_r \mid A) + P(E_r \mid B)}.$$

The common factor $\binom{n}{r}$ cancels. Hence,

$$\begin{aligned} P(A \mid E_r) &= \frac{3^r}{3^r + 3^{n-r}} \\ &= \boxed{\frac{1}{1 + 3^{n-2r}}}. \end{aligned}$$

Urn A is more likely than urn B when

$$P(A \mid E_r) > \frac{1}{2}.$$

This is equivalent to

$$3^r > 3^{n-r},$$

and hence

$$\boxed{r > \frac{n}{2}.}$$

Thus, more red than blue observations favor urn A, while more blue than red observations favor urn B. If $r = n/2$, the two urns remain equally likely.

4. Jenny owns an IPL ticket franchise with sales outlets at two locations. During a fixed booking period, let X and Y denote the numbers of tickets sold in Mumbai and Jaipur, respectively. Assume that X and Y are independent, with

$$X, Y \sim \mathcal{N}(70, 8^2).$$

Jenny is interested in the total ticket sales S, as well as the excess D of the sales in Mumbai over those in Jaipur, as a measure of the two outlets' relative marketing performance.

(a) (6 marks) For arbitrary u and v, evaluate

$$E\left[e^{uS+vD}\right].$$

Simplify your answer. State any assumptions you make.

Solution:

Multiple solution methods are possible.

- Since $S = X + Y$ and $D = X - Y$,

$$uS + vD = (u+v)X + (u-v)Y.$$

By the independence of X and Y,

$$\begin{aligned} \mathbb{E}\left[e^{uS+vD}\right] &= \mathbb{E}\left[e^{(u+v)X+(u-v)Y}\right] \\ &= \mathbb{E}\left[e^{(u+v)X}\right] \mathbb{E}\left[e^{(u-v)Y}\right]. \end{aligned}$$

For a normal random variable $Z \sim \mathcal{N}(\mu, \sigma^2)$,

$$\mathbb{E}[e^{tZ}] = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right).$$

Since $X, Y \sim \mathcal{N}(70, 8^2)$,

$$\begin{aligned}\mathbb{E}\left[e^{uS+vD}\right] &= \exp\left(70(u+v) + 32(u+v)^2\right) \\ &\quad \times \exp\left(70(u-v) + 32(u-v)^2\right) \\ &= \exp\left(140u + 32\left[(u+v)^2 + (u-v)^2\right]\right).\end{aligned}$$

Using

$$(u+v)^2 + (u-v)^2 = 2u^2 + 2v^2,$$

we obtain

$$\boxed{\mathbb{E}\left[e^{uS+vD}\right] = \exp\left(140u + 64u^2 + 64v^2\right).}$$

This holds for all real u and v, using the given independence and normality of X and Y.

- An alternate method is to write $W = uS + vD = (u+v)X + (u-v)Y$. Now we observe that W is normal since it is a sum of two normals. We therefore work out the expectaiton of $E[W] = 140u$, and the $Var(W) = 128(u^2 + v^2)$. Now we have derived the MGF of normal. From this we can directly write $E[e^W] = \exp(\mu_w + 0.5\sigma_W^2$. This route does not require observing that S and D are independent.
- A third possibility is to look at the joint distribution $f_{X,Y}(x, y)$ which is a classic 2D Gaussian. Now $uS+vD$ are functions of this random variable, and so is e^{uS+vD}. Therefore one can write the expectation from first principles. This is long but work.s

(b) (3 marks) Show steps and determine $\mathrm{Cov}(S, D)$.

Solution: Since

$$S = X + Y, \qquad D = X - Y,$$

we have

$$\begin{aligned}\mathrm{Cov}(S, D) &= \mathrm{Cov}(X+Y, X-Y) \\ &= \mathrm{Var}(X) - \mathrm{Cov}(X, Y) + \mathrm{Cov}(Y, X) - \mathrm{Var}(Y).\end{aligned}$$

Since X and Y are independent,

$$\mathrm{Cov}(X, Y) = 0.$$

Also,

$$\mathrm{Var}(X) = \mathrm{Var}(Y) = 8^2 = 64.$$

Therefore,

$$\begin{aligned}\mathrm{Cov}(S, D) &= 64 - 64 \\ &= \boxed{0}.\end{aligned}$$

There are many alternate routes

- Use Cov(S,D) = E[SD] - E[S]E[D]
- Write formula for Var(S+D)
- Observe independence of S and D in part(a)

(c) (4 marks) Given that more than 150 tickets are sold in total, find the probability that the better-performing outlet sells at least 8 more tickets. Unjustified assumptions receive no credit.

Solution:

The better-performing outlet sells at least 8 more tickets precisely when

$$|X - Y| \geq 8,$$

or equivalently,

$$|D| \geq 8.$$

Thus, the required probability is

$$P(|D| \geq 8 \mid S > 150).$$

The covariance is zero, so we suspect independance of S and D, though we are not sure. Consider the joint distribution

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-70)^2 + (y-70)^2}{2\sigma^2}\right).$$

$$F_{S,D}(s_0, d_0) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathbf{1}_{[x+y\leq s_0]}\, \mathbf{1}_{[x-y\leq d_0]}\, f_X(x) f_Y(y)\, dx\, dy.$$

Substitute $u = x + y$ (fix y; $x = u - y$, $dx = du$):

$$F_{S,D}(s_0, d_0) = \int_{-\infty}^{\infty}\int_{-\infty}^{s_0} \mathbf{1}[u - 2y \leq d_0]\, f_X(u-y) f_Y(y)\, du\, dy.$$

Substitute $v = u - 2y$ (fix u; $y = \frac{u-v}{2}$, $|dy| = \frac{1}{2}dv$):

$$F_{S,D}(s_0, d_0) = \int_{-\infty}^{s_0}\int_{-\infty}^{d_0} \tfrac{1}{2}\, f_X\left(\tfrac{u+v}{2}\right) f_Y\left(\tfrac{u-v}{2}\right)\, dv\, du.$$

Differentiating in s_0, d_0:

$$f_{S,D}(s,d) = \frac{1}{2}\, f_X\left(\frac{s+d}{2}\right) f_Y\left(\frac{s-d}{2}\right).$$

With $s' = s - 140$:

$$\left(\frac{s+d}{2} - 70\right)^2 + \left(\frac{s-d}{2} - 70\right)^2 = \left(\frac{s'+d}{2}\right)^2 + \left(\frac{s'-d}{2}\right)^2 = \frac{s'^2 + d^2}{2}.$$

So

$$f_{S,D}(s,d) = \frac{1}{2} \cdot \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(s-140)^2 + d^2}{4\sigma^2}\right) = \frac{1}{4\pi\sigma^2} \exp\left(-\frac{(s-140)^2 + d^2}{4\sigma^2}\right).$$

$$f_{S,D}(s,d) = \underbrace{\frac{1}{\sqrt{4\pi\sigma^2}} \exp\left(-\frac{(s-140)^2}{4\sigma^2}\right)}_{g(s)} \cdot \underbrace{\frac{1}{\sqrt{4\pi\sigma^2}} \exp\left(-\frac{d^2}{4\sigma^2}\right)}_{h(d)}.$$

Since $f_{S,D}(s,d) = g(s)h(d)$ factors into a function of s alone times a function of d alone, S and D are independent, with

$$S \sim N(140,\, 2\sigma^2) = N(140, 128), \qquad D \sim N(0,\, 2\sigma^2) = N(0, 128).$$

Alternatively, from part (a),

$$\mathbb{E}\left[e^{uS+vD}\right] = \exp(140u + 64u^2)\exp(64v^2).$$

This is the product of the moment-generating functions of S and D. Hence S and D are independent.

Therefore,

$$P(|D| \geq 8 \mid S > 150) = P(|D| \geq 8).$$

Since $D = X - Y$, and X and Y are independent normal random variables,

$$D \sim \mathcal{N}(0, 128).$$

Consequently,

$$\begin{aligned} P(|D| \geq 8) &= P\left(|Z| \geq \frac{8}{8\sqrt{2}}\right) \\ &= P\left(|Z| \geq \frac{1}{\sqrt{2}}\right) \\ &= 2\left[1 - \Phi\left(\frac{1}{\sqrt{2}}\right)\right], \end{aligned}$$

where $Z \sim \mathcal{N}(0,1)$ and Φ denotes its CDF.

Thus,

$$\boxed{P(|D| \geq 8 \mid S > 150) = 2\left[1 - \Phi\left(\frac{1}{\sqrt{2}}\right)\right] \approx 0.480.}$$

5. Let $X_1, \ldots, X_n$ be independent Bernoulli random variables with parameter p. Define

$$S_n = \sum_{i=1}^{n} X_i.$$

(a) (6 marks) For any real number a, show that for every $t > 0$,

$$P(S_n \geq a) \leq e^{-ta}(1 - p + pe^t)^n.$$

Solution: For each i,

$$\begin{aligned} M_{X_i}(t) &= \mathbb{E}(e^{tX_i}) \\ &= (1-p) + pe^t. \end{aligned}$$

Since the X_i are independent,

$$\begin{aligned} M_{S_n}(t) &= \mathbb{E}\left(e^{t\sum_{i=1}^{n} X_i}\right) \\ &= \prod_{i=1}^{n} \mathbb{E}(e^{tX_i}) \\ &= (1 - p + pe^t)^n. \end{aligned}$$

Hence,

$$\boxed{M_{S_n}(t) = (1 - p + pe^t)^n.}$$

For $t > 0$, the event $S_n \geq a$ implies

$$e^{tS_n} \geq e^{ta}.$$

Therefore, by Markov's inequality,

$$\begin{aligned} P(S_n \geq a) &= P(e^{tS_n} \geq e^{ta}) \\ &\leq e^{-ta}\mathbb{E}(e^{tS_n}). \end{aligned}$$

Using the expression for $M_{S_n}(t)$ obtained above,

$$\boxed{P(S_n \geq a) \leq e^{-ta}(1-p+pe^t)^n.}$$

(b) (6 marks) Now suppose $np < a < n$, and write $q = \frac{a}{n}$, $\quad p < q < 1$. Show in detail that

$$P(S_n \geq nq) \leq \left[\left(\frac{p}{q}\right)^q \left(\frac{1-p}{1-q}\right)^{1-q}\right]^n.$$

Solution: Since $a = nq$, the bound in part (a) is

$$B(t) = e^{-nqt}(1-p+pe^t)^n.$$

It is enough to minimize its logarithm:

$$\log B(t) = -nqt + n\log(1-p+pe^t).$$

Differentiating,

$$\frac{d}{dt}\log B(t) = -nq + n\frac{pe^t}{1-p+pe^t}.$$

At the minimum,

$$q = \frac{pe^t}{1-p+pe^t}.$$

Hence,

$$q(1-p) = pe^t(1-q),$$

so

$$e^{t^*} = \frac{q(1-p)}{p(1-q)}.$$

Thus,

$$\boxed{t^* = \log\left(\frac{q(1-p)}{p(1-q)}\right).}$$

Since $q > p$, we have $t^* > 0$.
Also,

$$\frac{d^2}{dt^2}\log B(t) = \frac{np(1-p)e^t}{(1-p+pe^t)^2} > 0.$$

Hence $\log B(t)$ is strictly convex, so t^* gives the minimum of the bound over $t > 0$.
Substituting this value into the bound,

$$1-p+pe^{t^*} = \frac{1-p}{1-q},$$

and

$$e^{-qt^*} = \left(\frac{p(1-q)}{q(1-p)}\right)^q.$$

Therefore,

$$P(S_n \geq nq) \leq \left[\left(\frac{p(1-q)}{q(1-p)}\right)^q \frac{1-p}{1-q}\right]^n$$
$$= \left[\left(\frac{p}{q}\right)^q \left(\frac{1-p}{1-q}\right)^{1-q}\right]^n.$$

Hence,

$$\boxed{P(S_n \geq nq) \leq \left[\left(\frac{p}{q}\right)^q \left(\frac{1-p}{1-q}\right)^{1-q}\right]^n.}$$

6. A stick of unit length is broken at a point chosen uniformly at random along its length. Let U denote the length of the piece containing the left endpoint. Define $X = U(1-U)$.

   (a) (3 marks) Determine $\mathrm{Var}(X)$. Show three intermediate steps.

   Solution:

   Since

   $$X = U(1-U) = U - U^2,$$

   we have

   $$\mathbb{E}(X) = \mathbb{E}(U) - \mathbb{E}(U^2).$$

   For $U \sim \mathrm{Uniform}(0,1)$,

   $$\mathbb{E}(U) = \frac{1}{2}, \qquad \mathbb{E}(U^2) = \frac{1}{3}.$$

   Therefore,

   $$\boxed{\mathbb{E}(X) = \frac{1}{6}.}$$

   Next,

   $$X^2 = U^2(1-U)^2 = U^2 - 2U^3 + U^4.$$

   Hence,

   $$\mathbb{E}(X^2) = \mathbb{E}(U^2) - 2\mathbb{E}(U^3) + \mathbb{E}(U^4)$$
   $$= \frac{1}{3} - \frac{1}{2} + \frac{1}{5}$$
   $$= \frac{1}{30}.$$

   Therefore,

   $$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$$
   $$= \frac{1}{30} - \frac{1}{36}$$
   $$= \frac{1}{180}.$$

   Thus,

   $$\boxed{\mathbb{E}(X) = \frac{1}{6}, \qquad \mathrm{Var}(X) = \frac{1}{180}.}$$

   (b) (3 marks) Determine $\mathrm{Cov}(U, X)$. Show three intermediate steps.

Solution: We have

$$UX = U^2(1-U) = U^2 - U^3.$$

Therefore,

$$\begin{aligned}\mathbb{E}(UX) &= \mathbb{E}(U^2) - \mathbb{E}(U^3)\\ &= \frac{1}{3} - \frac{1}{4}\\ &= \frac{1}{12}.\end{aligned}$$

Using part (a),

$$\mathbb{E}(U)\mathbb{E}(X) = \frac{1}{2}\cdot\frac{1}{6} = \frac{1}{12}.$$

Hence,

$$\boxed{\mathrm{Cov}(U, X) = 0.}$$

(c) (3 marks) Claim: U and X are NOT independent. Do you agree? Justify quantitatively.

Solution: However, U and X are not independent. For example,

$$X > \frac{3}{16}$$

is equivalent to

$$U(1-U) > \frac{3}{16},$$

which gives

$$\frac{1}{4} < U < \frac{3}{4}.$$

Thus,

$$P\left(X > \frac{3}{16}\right) = \frac{1}{2}.$$

On the other hand,

$$P\left(X > \frac{3}{16}\,\middle|\, U < \frac{1}{4}\right) = 0.$$

Therefore,

$$P\left(X > \frac{3}{16}\,\middle|\, U < \frac{1}{4}\right) \neq P\left(X > \frac{3}{16}\right),$$

and hence U and X are not independent.

Thus,

$$\boxed{U \text{ and } X \text{ are uncorrelated, but not independent.}}$$

(d) (6 marks) Let

$$Y = \begin{cases} 1, & U < \frac{1}{2},\\ 0, & U \geq \frac{1}{2}.\end{cases}$$

Show that X and Y are independent.

Solution: Clearly,

$$P(Y = 1) = P\left(U < \frac{1}{2}\right) = \frac{1}{2},$$

and similarly,

$$P(Y = 0) = \frac{1}{2}.$$

Also,

$$0 \leq X \leq \frac{1}{4}.$$

Let $0 \leq x < \frac{1}{4}$. We consider the event.

$$X \leq x.$$

Solving

$$U(1 - U) \leq x$$

gives

$$U^2 - U + x \geq 0.$$

The two roots are

$$a = \frac{1 - \sqrt{1 - 4x}}{2}$$

and

$$1 - a = \frac{1 + \sqrt{1 - 4x}}{2}.$$

Hence,

$$\{X \leq x\} = \{U \leq a\} \cup \{U \geq 1 - a\}.$$

Since U is uniform on $(0, 1)$,

$$P(X \leq x) = a + a = 2a.$$

Now,

$$\{X \leq x,\ Y = 1\} = \{U \leq a\},$$

because $Y = 1$ corresponds to $U < 1/2$.
Therefore,

$$P(X \leq x,\ Y = 1) = a.$$

But

$$P(X \leq x)P(Y = 1) = (2a)\left(\frac{1}{2}\right) = a.$$

Thus,

$$P(X \leq x,\ Y = 1) = P(X \leq x)P(Y = 1).$$

Similarly,

$$P(X \leq x,\ Y = 0) = P(X \leq x)P(Y = 0).$$

For $x < 0$, both $P(X \leq x)$ and $P(X \leq x, Y = y)$ are zero. For $x \geq \frac{1}{4}$, we have $P(X \leq x) = 1$, and hence

$$P(X \leq x, Y = y) = P(Y = y) = P(X \leq x)P(Y = y),$$

for $y = 0, 1$.
Hence the distribution of X is unchanged by the value of Y, and therefore

$$\boxed{X \text{ and } Y \text{ are independent.}}$$

7. Let $X_1, \ldots, X_n$ be i.i.d. random variables drawn from a continuous probability distribution symmetric

about 0. Suppose that

$$\mathbb{E}(|X_1|) = \mu, \qquad \mathbb{E}(X_1^2) = \sigma^2 < \infty.$$

Define

$$S = \sum_{i=1}^{n} X_i$$

and let N denote the number of positive observations among $X_1, \ldots, X_n$.

(a) (2 marks) Determine $\mathbb{E}(S)$. Show two intermediate steps.

Solution: Since the distribution is continuous and symmetric about 0,

$$P(X_i > 0) = P(X_i[le0) = \frac{1}{2}.$$

Linearity and symmetry therefore has

$$\boxed{\mathbb{E}(S) = 0}$$

(b) (3 marks) Determine the distribution of N. Show two intermediate steps.

Solution:

$$N = \sum_{i=1}^{n} \mathbf{1}_{\{X_i > 0\}},$$

Since the distribution is continuous and symmetric about 0,

$$P(X_i > 0) = P(X_i \leq 0) = \frac{1}{2}.$$

The random variables $\mathbf{1}_{\{X_i>0\}}$ are independent, have only two possible values as outcome, and have identical probability, hence

$$\boxed{N \sim \text{Binomial}\left(n, \frac{1}{2}\right).}$$

(c) (5 marks) Determine $\text{Cov}(S, N)$. Show three intermediate steps.

Solution:

Since $\mathbb{E}(S) = 0$,

$$\text{Cov}(S, N) = \mathbb{E}(SN).$$

Expanding,

$$\mathbb{E}(SN) = \sum_{i=1}^{n}\sum_{j=1}^{n} \mathbb{E}\left[X_i \mathbf{1}_{\{X_j>0\}}\right].$$

Next, If $i \neq j$, independence between X_i and X_j results in

$$\mathbb{E}\left[X_i \mathbf{1}_{\{X_j>0\}}\right] = \mathbb{E}(X_i)\mathbb{E}[\mathbf{1}_{X_j>0)}]$$

and

$$\mathbb{E}\left[X_i \mathbf{1}_{\{X_j>0\}}\right] = \mathbb{E}(X_i)P(X_j > 0) = 0.$$

Thus,

$$\text{Cov}(S, N) = n\,\mathbb{E}\left[X_1 \mathbf{1}_{\{X_1>0\}}\right].$$

So far, the given value of mod X has not yet entered the picture. We note

$$\mu = E[|X_1|] = \int_0^\infty t f|X_1|(t)dt = \int_0^\infty t(2fX_1(t))dt$$

Meanwhile, the term needed for the covariance is $E[X_1\mathbf{1}_{X1>0}] = \int_0^\infty t f_{X_1}(t)dt$.

$$\mathbb{E}(|X_1|) = 2\mathbb{E}\left[X_1\mathbf{1}_{\{X_1>0\}}\right].$$

Since $\mathbb{E}(|X_1|) = \mu$,

$$\mathbb{E}\left[X_1\mathbf{1}_{\{X_1>0\}}\right] = \frac{\mu}{2}.$$

Hence,

$$\boxed{\mathrm{Cov}(S,N) = \frac{n\mu}{2}.}$$

(d) (4 marks) For $k = 0, 1, \ldots, n$, determine $\mathbb{E}(S \mid N = k)$. Show three intermediate steps.

Solution: We look for mean of a (simpler) observation known to be positive, i.e., the conditional expectation, $\mathbb{E}[X_1|X_1 > 0]$.
From part (c),

$$\mathbb{E}\left[X_1\mathbf{1}_{\{X_1>0\}}\right] = \frac{\mu}{2}.$$

Since $P(X_1 > 0) = 1/2$, using standard ideas from conditional probability

$$\mathbb{E}(X_1 \mid X_1 > 0) = \frac{\mathbb{E}\left[X_1\mathbf{1}_{\{X_1>0\}}\right]}{P(X_1 > 0)} = \mu.$$

By symmetry,

$$\mathbb{E}(X_1 \mid X_1 < 0) = -\mu.$$

Given $N = k$ each of the k positive observations has conditional mean μ, and each of the $n - k$ negative observation has conditional mean $-\mu$. Therefore,

$$\begin{aligned}\mathbb{E}(S \mid N = k) &= k\mu - (n-k)\mu \\ &= (2k - n)\mu.\end{aligned}$$

Thus,

$$\boxed{\mathbb{E}(S \mid N = k) = (2k - n)\mu.}$$

(e) (6 marks) Determine $\mathrm{Var}(S \mid N = k)$. Show three intermediate steps.

Solution: By symmetry,

$$\mathbb{E}(X_1^2 \mid X_1 > 0) = \mathbb{E}(X_1^2 \mid X_1 < 0).$$

Also,

$$\sigma^2 = \mathbb{E}(X_1^2) = \frac{1}{2}\mathbb{E}(X_1^2 \mid X_1 > 0) + \frac{1}{2}\mathbb{E}(X_1^2 \mid X_1 < 0).$$

Hence,

$$\mathbb{E}(X_1^2 \mid X_1 > 0) = \mathbb{E}(X_1^2 \mid X_1 < 0) = \sigma^2.$$

Using the conditional means obtained in part (d),

$$\mathrm{Var}(X_1 \mid X_1 > 0) = \sigma^2 - \mu^2,$$

and similarly,

$$\mathrm{Var}(X_1 \mid X_1 < 0) = \sigma^2 - \mu^2.$$

Let A specify which k observations are positive. Once A is fixed, we are conditioning each observation on its own sign. Since the original observations are independent, they remain independent under this conditioning. Thus, for every such choice A,

$$\mathrm{Var}(S \mid A) = k(\sigma^2 - \mu^2) + (n-k)(\sigma^2 - \mu^2) = n(\sigma^2 - \mu^2).$$

Also, for every such choice A, the conditional mean is the same from the previou step. Averaging over these patterns, which together make up the event $\{N = k\}$, gives

$$\boxed{\mathrm{Var}(S \mid N = k) = n(\sigma^2 - \mu^2).}$$

8. Let X be a random variable satisfying

$$\mathbb{E}[X] = \mu > 0, \qquad \mathbb{E}[X^2] = m_2 < \infty.$$

(a) (2 marks) Justify and apply Jensen's inequality to show that $m_2 \geq \mu^2$.

Solution:

The function

$$f(x) = x^2$$

is convex. Therefore, by Jensen's inequality,

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

Hence,

$$(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2].$$

Using the given values,

$$\boxed{\mu^2 \leq m_2.}$$

(b) (6 marks) Let $0 < \theta < 1$. Show that

$$P(X \leq \theta\mu) \leq \frac{m_2 - \mu^2}{(1-\theta)^2\mu^2}.$$

Solution:

Let $a = (1-\theta)\mu > 0$. Then

$$\begin{aligned} P(X \leq \theta\mu) &= P(\mu - X \geq a) \\ &\leq P\big((X-\mu)^2 \geq a^2\big) \qquad \text{since both sides positive} \end{aligned}$$

Since $(X-\mu)^2$ is nonnegative, Markov's inequality gives

$$\begin{aligned} P(X \leq \theta\mu) &\leq \frac{\mathbb{E}[(X-\mu)^2]}{(1-\theta)^2\mu^2} \\ &= \frac{m_2 - \mu^2}{(1-\theta)^2\mu^2}. \end{aligned}$$

Hence,

$$\boxed{P(X \leq \theta\mu) \leq \frac{m_2 - \mu^2}{(1-\theta)^2\mu^2}.}$$

A slight variation can be obtained using Chebyshev's inequality arguing on the modulus instead of the square. Let $a = (1-\theta)\mu > 0$. Since $\theta\mu = \mu - a$,

$$\begin{aligned} P(X \le \theta\mu) &= P(\mu - X \ge a) \\ &\le P\big((\mu - X)^2 \ge a^2\big) \\ &= P(|X - \mu| \ge a) \\ &\le \frac{\operatorname{Var}(X)}{a^2} = \frac{m_2 - \mu^2}{(1-\theta)^2\mu^2}. \end{aligned}$$

The last inequality is Chebyshev's inequality.

(c) (10 bonus marks) The Cauchy–Schwarz inequality can be stated as

$$\big|\mathbb{E}[YZ]\big| \le \sqrt{\mathbb{E}[Y^2]\,\mathbb{E}[Z^2]}.$$

Using the Cauchy–Schwarz inequality, show that

$$P(X > \theta\mu) \ge \frac{(1-\theta)^2\mu^2}{m_2}.$$

Teaching-team notes. This bonus question should be skipped in the first pass. It is intended only for someone who has completed all other questions.

Solution: We want to bound a probability using a given inequality about expectations. For an event A, its indicator satisfies

$$\mathbb{E}[\mathbf{1}_A^2] = \mathbb{E}[\mathbf{1}_A] = P(A).$$

This suggests applying Cauchy–Schwarz to X and $\mathbf{1}_A$. Let

$$A = \{X > \theta\mu\}.$$

We write

$$\mathbb{E}[X] = \mathbb{E}[X\mathbf{1}_A] + \mathbb{E}[X\mathbf{1}_{A^c}].$$

On A^c,

$$X \le \theta\mu.$$

Since $X \le \theta\mu$ on A^c,

$$\mathbb{E}[X\mathbf{1}_{A^c}] \le \theta\mu P(A^c) \le \theta\mu.$$

Therefore,

$$\mu \le \mathbb{E}[X\mathbf{1}_A] + \theta\mu,$$

and hence

$$(1-\theta)\mu \le \mathbb{E}[X\mathbf{1}_A].$$

By the Cauchy-Schwarz inequality,

$$\mathbb{E}[X\mathbf{1}_A] \le \sqrt{\mathbb{E}[X^2]\mathbb{E}[\mathbf{1}_A^2]}.$$

Since

$$\mathbf{1}_A^2 = \mathbf{1}_A,$$

we have

$$\mathbb{E}[\mathbf{1}_A^2] = P(A).$$

Thus,

$$(1-\theta)\mu \leq \sqrt{m_2 P(A)}.$$

Squaring both sides,

$$(1-\theta)^2\mu^2 \leq m_2 P(A).$$

Therefore,

$$\boxed{P(X > \theta\mu) \geq \frac{(1-\theta)^2\mu^2}{m_2}.}$$